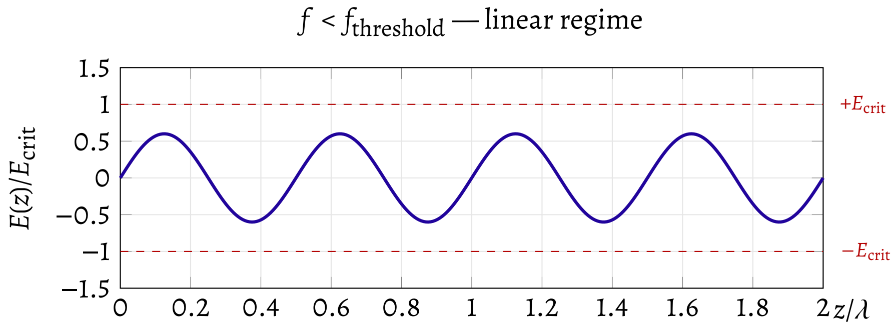
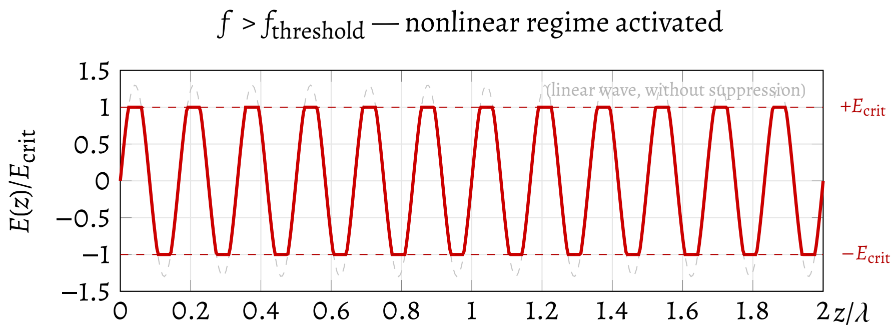


Field never reaches E_{crit} · No semi-particle formed · No electron ejected

(a) Linear regime: $f < f_{\text{threshold}}$ (low frequency)



Field reaches $E_{\text{crit}} \Rightarrow$ nonlinear suppression · Semi-particles formed at the peaks · Electrons ejected

(b) Nonlinear regime: $f > f_{\text{threshold}}$ (high frequency)

Figure 10 – Two regimes of the photoelectric effect under the hypothesis of restitutive nonlinearity (cf. Appendix R). **(a)** For $f < f_{\text{threshold}}$, the amplitude remains below E_{crit} (dashed red): linear regime, with no semi-particles and no ejection of electrons, regardless of intensity. **(b)** For $f > f_{\text{threshold}}$, the field would reach amplitudes above E_{crit} (gray ghost), but the saturation limits the peaks (solid red curve), forming localized excitations (semi-particles) that transfer concentrated energy to the electrons.